LAPORAN PRAKTIKUM PRA-UTS

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JURUSAN TEKNIK INFORMATIKA

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BAB I
PENDAHULUAN

1. 1 Latar Belakang

Perkembangan teknologi informasi yang semakin pesat mendorong berbagai sektor usaha, termasuk toko buku, untuk memanfaatkan sistem berbasis database dalam mengelola data. Pengelolaan data secara manual memiliki banyak kelemahan, seperti risiko kesalahan pencatatan, kesulitan dalam pencarian data, serta kurang efisien dalam proses transaksi.

Dengan adanya database, pengelolaan data buku dan transaksi penjualan dapat dilakukan secara lebih terstruktur, cepat, dan akurat. Sistem database memungkinkan penyimpanan data dalam jumlah besar, serta memudahkan proses pengolahan seperti pencarian, pengurutan, dan pembaruan data.

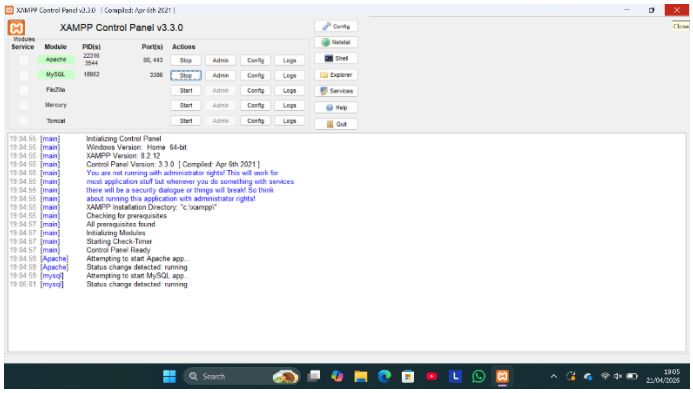
Pada praktikum ini, dilakukan pembuatan database sederhana untuk toko buku yang terdiri dari tabel buku dan tabel transaksi. Melalui praktikum ini, diharapkan mahasiswa dapat memahami konsep dasar database serta mampu mengimplementasikan perintah SQL seperti SELECT, WHERE, ORDER BY, UPDATE, dan DELETE dalam pengelolaan data.

1.2 Tujuan Praktikum

1. Membuat database dan tabel.
2. Menentukan dan menggunakan PRIMARY KEY pada setiap tabel.
3. Menginput data ke dalam tabel menggunakan perintah INSERT.
4. Menampilkan data menggunakan perintah SELECT.
5. Menggunakan kondisi WHERE untuk menyaring data.
6. Mengurutkan data menggunakan ORDER BY (ASC dan DESC).
7. Melakukan perubahan data menggunakan perintah UPDATE.
8. Menghapus data menggunakan perintah DELETE sesuai kondisi tertentu.

BAB II
HASIL PRAKTIKUM

2.1 Mengakses MySQL Melalui Command Prompt




```
C:\Users\tsbta>cd/xampp/mysql/bin

C:\xampp\mysql\bin>mysql -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 22
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

2.2 Melihat Database Yang Sudah Dibuat

```
MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| akademik           |
| information_schema |
| mysql              |
| performance_schema |
| phpmyadmin         |
| pmb                |
| test               |
| toko buku          |
+--------------------+
8 rows in set (0.001 sec)
```

2.3 Membuat Database dan Tabel

```
MariaDB [(none)]> create database bookstore;
Query OK, 1 row affected (0.002 sec)
```

```
MariaDB [(none)]> use bookstore;
Database changed
```

```
MariaDB [bookstore]> create table buku(
    -> id int primary key,
    -> judul_buku varchar(40),
    -> nama_penulis varchar(40),
    -> penerbit varchar(30),
    -> harga int(11));
Query OK, 0 rows affected (0.017 sec)
```

```
MariaDB [bookstore]> desc buku;
+-------------+------------+------+-----+---------+-------+
| Field       | Type       | Null | Key | Default | Extra |
+-------------+------------+------+-----+---------+-------+
| id          | int(11)    | NO   | PRI | NULL    |       |
| judul_buku  | varchar(40)| YES  |     | NULL    |       |
| nama_penulis| varchar(40)| YES  |     | NULL    |       |
| penerbit    | varchar(30)| YES  |     | NULL    |       |
| harga       | int(11)    | YES  |     | NULL    |       |
+-------------+------------+------+-----+---------+-------+
5 rows in set (0.030 sec)
```

```
MariaDB [bookstore]> create table transaksi(
    -> id_transaksi int primary key,
    -> judul_buku varchar(40),
    -> jumlah_buku varchar(10),
    -> total_harga varchar(20),
    -> tanggal_transaksi varchar(10));
Query OK, 0 rows affected (0.016 sec)
```

```
MariaDB [bookstore]> desc transaksi;
+------------------+------------+------+-----+---------+-------+
| Field            | Type       | Null | Key | Default | Extra |
+------------------+------------+------+-----+---------+-------+
| id_transaksi     | int(11)    | NO   | PRI | NULL    |       |
| judul_buku       | varchar(40)| YES  |     | NULL    |       |
| jumlah_buku      | varchar(10)| YES  |     | NULL    |       |
| total_harga      | varchar(20)| YES  |     | NULL    |       |
| tanggal_transaksi| varchar(10)| YES  |     | NULL    |       |
+------------------+------------+------+-----+---------+-------+
5 rows in set (0.021 sec)
```

```
MariaDB [bookstore]> show tables;
+--------------------+
| Tables_in_bookstore |
+--------------------+
| buku               |
| transaksi          |
+--------------------+
2 rows in set (0.001 sec)
```

2.4 Mengisi Data Tabel

```
MariaDB [bookstore]> insert into buku values
    -> (101,'Jakarta Sebelum Pagi','Ziggy Z','Grasindo','Rp85.000,00'),
    -> (102,'Teka-Teki Rumah Aneh','Uketsu','Gramedia','Rp60.000,00'),
    -> (103,'Tragedi Pedang Keadilan','Keigo Higashino','Gramedia','Rp90.000,00'),
    -> (104,'Bumi Manusia','Pramoedya Ananta Toer','Lentera Dipantara','Rp180.000,00'),
    -> (105,'Laskar Pelangi','Andrea Hinata','Bentang Pustaka','Rp75.000,00'),
    -> (106,'Aroma Karsa','Dee Lestari','Bentang Pustaka','Rp160.000,00'),
    -> (107,'Ronggeng Dukuh Paruk','Ahmad Tohari','Gramedia','Rp90.000,00'),
    -> (108,'Gadis Kretek','Ratih Kumala','Gramedia','Rp70.000,00'),
    -> (109,'Tiga Dalam Kayu','Ziggy Z','Grasindo','Rp50.000,00'),
    -> (110,'Laut Bercerita','Leila S. Chudori','Gramedia','Rp85.000,00');
Query OK, 10 rows affected, 10 warnings (0.012 sec)
Records: 10  Duplicates: 0  Warnings: 10
```

```
MariaDB [bookstore]> select * from buku;
+-----+--------------------------+----------------------+--------------------+--------+
| id  | judul_buku               | nama_penulis         | penerbit           | harga  |
+-----+--------------------------+----------------------+--------------------+--------+
| 101 | Jakarta Sebelum Pagi     | Ziggy Z              | Grasindo           |      0 |
| 102 | Teka-Teki Rumah Aneh     | Uketsu               | Gramedia           |      0 |
| 103 | Tragedi Pedang Keadilan  | Keigo Higashino      | Gramedia           |      0 |
| 104 | Bumi Manusia             | Pramoedya Ananta Toer | Lentera Dipantara |      0 |
| 105 | Laskar Pelangi           | Andrea Hinata        | Bentang Pustaka    |      0 |
| 106 | Aroma Karsa              | Dee Lestari          | Bentang Pustaka    |      0 |
| 107 | Ronggeng Dukuh Paruk     | Ahmad Tohari         | Gramedia           |      0 |
| 108 | Gadis Kretek             | Ratih Kumala         | Gramedia           |      0 |
| 109 | Tiga Dalam Kayu          | Ziggy Z              | Grasindo           |      0 |
| 110 | Laut Bercerita           | Leila S. Chudori     | Gramedia           |      0 |
+-----+--------------------------+----------------------+--------------------+--------+
10 rows in set (0.001 sec)
```

```
MariaDB [bookstore]> insert into transaksi values
    -> (1,'Gadis Kretek, Laskar Pelangi','2','Rp145.000,00','1 Mei 2026'),
    -> (2,'Bumi Manusia','1','Rp180.000,00','1 Mei 2026'),
    -> (3,'Aroma Karsa, Tiga Dalam Kayu, Laut Bercerita','3','Rp295.000,00','1 Mei 2026'),
    -> (4,'Jakarta Sebelum Pagi','1','Rp85.000,00','1 Mei 2026'),
    -> (5,'Aroma Karsa','1','Rp160.000,00','2 Mei 2026'),
    -> (6,'Tragedi Pedang Keadilan','1','Rp90.000,00','2 Mei 2026'),
    -> (7,'Ronggeng Dukuh Paruk, Laut Bercerita','2','Rp175.000,00','2 Mei 2026'),
    -> (8,'Tiga Dalam Kayu','1','Rp50.000,00','2 Mei 2026'),
    -> (9,'Laskar Pelangi','1','Rp75.000,00','3 Mei 2026'),
    -> (10,'Teka-Teki Rumah Aneh','1','Rp60.000,00','3 Mei 2026');
Query OK, 10 rows affected, 1 warning (0.011 sec)
Records: 10  Duplicates: 0  Warnings: 1
```

```
MariaDB [bookstore]> select * from transaksi;
+--------------+-----------------------------------------+-------------+--------------+-------------------+
| id_transaksi | judul_buku                              | jumlah_buku | total_harga  | tanggal_transaksi |
+--------------+-----------------------------------------+-------------+--------------+-------------------+
|            1 | Gadis Kretek, Laskar Pelangi            | 2           | Rp145.000,00 | 1 Mei 2026        |
|            2 | Bumi Manusia                            | 1           | Rp180.000,00 | 1 Mei 2026        |
|            3 | Aroma Karsa, Tiga Dalam Kayu, Laut Berce | 3           | Rp295.000,00 | 1 Mei 2026        |
|            4 | Jakarta Sebelum Pagi                    | 1           | Rp85.000,00  | 1 Mei 2026        |
|            5 | Aroma Karsa                             | 1           | Rp160.000,00 | 2 Mei 2026        |
|            6 | Tragedi Pedang Keadilan                 | 1           | Rp90.000,00  | 2 Mei 2026        |
|            7 | Ronggeng Dukuh Paruk, Laut Bercerita    | 2           | Rp175.000,00 | 2 Mei 2026        |
|            8 | Tiga Dalam Kayu                         | 1           | Rp50.000,00  | 2 Mei 2026        |
|            9 | Laskar Pelangi                          | 1           | Rp75.000,00  | 3 Mei 2026        |
|           10 | Teka-Teki Rumah Aneh                    | 1           | Rp60.000,00  | 3 Mei 2026        |
+--------------+-----------------------------------------+-------------+--------------+-------------------+
10 rows in set (0.001 sec)
```

2.5 Menggunakan Perintah WHERE dan ORDER BY (ASC dan DESC)

```
MariaDB [bookstore]> select * from buku
    -> where penerbit='Gramedia';
+-----+--------------------------+------------------+----------+------+
| id  | judul_buku               | nama_penulis     | penerbit | harga |
+-----+--------------------------+------------------+----------+------+
| 102 | Teka-Teki Rumah Aneh     | Uketsu           | Gramedia |     0 |
| 103 | Tragedi Pedang Keadilan  | Keigo Higashino  | Gramedia |     0 |
| 107 | Ronggeng Dukuh Paruk     | Ahmad Tohari     | Gramedia |     0 |
| 108 | Gadis Kretek             | Ratih Kumala     | Gramedia |     0 |
| 110 | Laut Bercerita           | Leila S. Chudori | Gramedia |     0 |
+-----+--------------------------+------------------+----------+------+
5 rows in set (0.001 sec)

MariaDB [bookstore]> select * from transaksi
    -> where jumlah_buku='1';
+--------------+--------------------------+-------------+--------------+-------------------+
| id_transaksi | judul_buku               | jumlah_buku | total_harga  | tanggal_transaksi |
+--------------+--------------------------+-------------+--------------+-------------------+
|            2 | Bumi Manusia             | 1           | Rp180.000,00 | 1 Mei 2026        |
|            4 | Jakarta Sebelum Pagi     | 1           | Rp85.000,00  | 1 Mei 2026        |
|            5 | Aroma Karsa              | 1           | Rp160.000,00 | 2 Mei 2026        |
|            6 | Tragedi Pedang Keadilan  | 1           | Rp90.000,00  | 2 Mei 2026        |
|            8 | Tiga Dalam Kayu          | 1           | Rp50.000,00  | 2 Mei 2026        |
|            9 | Laskar Pelangi           | 1           | Rp75.000,00  | 3 Mei 2026        |
|           10 | Teka-Teki Rumah Aneh     | 1           | Rp60.000,00  | 3 Mei 2026        |
+--------------+--------------------------+-------------+--------------+-------------------+
7 rows in set (0.001 sec)
```

```
MariaDB [bookstore]> select * from buku
    -> order by judul_buku ASC;
+-----+--------------------------+-----------------------+--------------------+-------+
| id  | judul_buku               | nama_penulis          | penerbit           | harga |
+-----+--------------------------+-----------------------+--------------------+-------+
| 106 | Aroma Karsa              | Dee Lestari           | Bentang Pustaka    |     0 |
| 104 | Bumi Manusia             | Pramoedya Ananta Toer | Lentera Dipantara  |     0 |
| 108 | Gadis Kretek             | Ratih Kumala          | Gramedia           |     0 |
| 101 | Jakarta Sebelum Pagi     | Ziggy Z               | Grasindo           |     0 |
| 105 | Laskar Pelangi           | Andrea Hinata         | Bentang Pustaka    |     0 |
| 110 | Laut Bercerita           | Leila S. Chudori      | Gramedia           |     0 |
| 107 | Ronggeng Dukuh Paruk     | Ahmad Tohari          | Gramedia           |     0 |
| 102 | Teka-Teki Rumah Aneh     | Uketsu                | Gramedia           |     0 |
| 109 | Tiga Dalam Kayu          | Ziggy Z               | Grasindo           |     0 |
| 103 | Tragedi Pedang Keadilan  | Keigo Higashino       | Gramedia           |     0 |
+-----+--------------------------+-----------------------+--------------------+-------+
10 rows in set (0.001 sec)

MariaDB [bookstore]> select * from buku
    -> order by judul_buku DESC;
+-----+--------------------------+-----------------------+--------------------+-------+
| id  | judul_buku               | nama_penulis          | penerbit           | harga |
+-----+--------------------------+-----------------------+--------------------+-------+
| 103 | Tragedi Pedang Keadilan  | Keigo Higashino       | Gramedia           |     0 |
| 109 | Tiga Dalam Kayu          | Ziggy Z               | Grasindo           |     0 |
| 102 | Teka-Teki Rumah Aneh     | Uketsu                | Gramedia           |     0 |
| 107 | Ronggeng Dukuh Paruk     | Ahmad Tohari          | Gramedia           |     0 |
| 110 | Laut Bercerita           | Leila S. Chudori      | Gramedia           |     0 |
| 105 | Laskar Pelangi           | Andrea Hinata         | Bentang Pustaka    |     0 |
| 101 | Jakarta Sebelum Pagi     | Ziggy Z               | Grasindo           |     0 |
| 108 | Gadis Kretek             | Ratih Kumala          | Gramedia           |     0 |
| 104 | Bumi Manusia             | Pramoedya Ananta Toer | Lentera Dipantara  |     0 |
| 106 | Aroma Karsa              | Dee Lestari           | Bentang Pustaka    |     0 |
+-----+--------------------------+-----------------------+--------------------+-------+
10 rows in set (0.001 sec)
```

2.6 Update Data

```
MariaDB [bookstore]> update buku
    -> set harga=85000
    -> where id=101;
Query OK, 1 row affected (0.010 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [bookstore]> select * from buku;
+-----+--------------------------+-----------------------+--------------------+-------+
| id  | judul_buku               | nama_penulis          | penerbit           | harga |
+-----+--------------------------+-----------------------+--------------------+-------+
| 101 | Jakarta Sebelum Pagi     | Ziggy Z               | Grasindo           | 85000 |
| 102 | Teka-Teki Rumah Aneh     | Uketsu                | Gramedia           |     0 |
| 103 | Tragedi Pedang Keadilan  | Keigo Higashino       | Gramedia           |     0 |
| 104 | Bumi Manusia             | Pramoedya Ananta Toer | Lentera Dipantara  |     0 |
| 105 | Laskar Pelangi           | Andrea Hinata         | Bentang Pustaka    |     0 |
| 106 | Aroma Karsa              | Dee Lestari           | Bentang Pustaka    |     0 |
| 107 | Ronggeng Dukuh Paruk     | Ahmad Tohari          | Gramedia           |     0 |
| 108 | Gadis Kretek             | Ratih Kumala          | Gramedia           |     0 |
| 109 | Tiga Dalam Kayu          | Ziggy Z               | Grasindo           |     0 |
| 110 | Laut Bercerita           | Leila S. Chudori      | Gramedia           |     0 |
+-----+--------------------------+-----------------------+--------------------+-------+
10 rows in set (0.000 sec)
```

```
MariaDB [bookstore]> update buku
    -> set judul_buku='Teak-Teki Gambar Aneh', penerbit='Grasindo', harga=65000
    -> where id=102;
Query OK, 1 row affected (0.010 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [bookstore]> select * from buku;
+-----+--------------------------+-----------------------+--------------------+-------+
| id  | judul_buku               | nama_penulis          | penerbit           | harga |
+-----+--------------------------+-----------------------+--------------------+-------+
| 101 | Jakarta Sebelum Pagi     | Ziggy Z               | Grasindo           | 85000 |
| 102 | Teak-Teki Gambar Aneh    | Uketsu                | Grasindo           | 65000 |
| 103 | Tragedi Pedang Keadilan  | Keigo Higashino       | Gramedia           |     0 |
| 104 | Bumi Manusia             | Pramoedya Ananta Toer | Lentera Dipantara  |     0 |
| 105 | Laskar Pelangi           | Andrea Hinata         | Bentang Pustaka    |     0 |
| 106 | Aroma Karsa              | Dee Lestari           | Bentang Pustaka    |     0 |
| 107 | Ronggeng Dukuh Paruk     | Ahmad Tohari          | Gramedia           |     0 |
| 108 | Gadis Kretek             | Ratih Kumala          | Gramedia           |     0 |
| 109 | Tiga Dalam Kayu          | Ziggy Z               | Grasindo           |     0 |
| 110 | Laut Bercerita           | Leila S. Chudori      | Gramedia           |     0 |
+-----+--------------------------+-----------------------+--------------------+-------+
10 rows in set (0.001 sec)
```

2.7 Delete Data

```
MariaDB [bookstore]> delete from transaksi
    -> where id_transaksi=1;
Query OK, 1 row affected (0.009 sec)

MariaDB [bookstore]> select * from transaksi;
+--------------+-----------------------------------------+-------------+--------------+-------------------+
| id_transaksi | judul_buku                              | jumlah_buku | total_harga  | tanggal_transaksi |
+--------------+-----------------------------------------+-------------+--------------+-------------------+
|            2 | Bumi Manusia                            | 1           | Rp180.000,00 | 1 Mei 2026        |
|            3 | Aroma Karsa, Tiga Dalam Kayu, Laut Berce | 3           | Rp295.000,00 | 1 Mei 2026        |
|            4 | Jakarta Sebelum Pagi                    | 1           | Rp85.000,00  | 1 Mei 2026        |
|            5 | Aroma Karsa                             | 1           | Rp160.000,00 | 2 Mei 2026        |
|            6 | Tragedi Pedang Keadilan                 | 1           | Rp90.000,00  | 2 Mei 2026        |
|            7 | Ronggeng Dukuh Paruk, Laut Bercerita    | 2           | Rp175.000,00 | 2 Mei 2026        |
|            8 | Tiga Dalam Kayu                         | 1           | Rp50.000,00  | 2 Mei 2026        |
|            9 | Laskar Pelangi                          | 1           | Rp75.000,00  | 3 Mei 2026        |
|           10 | Teka-Teki Rumah Aneh                    | 1           | Rp60.000,00  | 3 Mei 2026        |
+--------------+-----------------------------------------+-------------+--------------+-------------------+
9 rows in set (0.000 sec)
```

```
MariaDB [bookstore]> delete from transaksi
    -> where id_transaksi=2 or id_transaksi=3;
Query OK, 2 rows affected (0.008 sec)

MariaDB [bookstore]> select * from transaksi;
+--------------+-----------------------------------------+-------------+--------------+-------------------+
| id_transaksi | judul_buku                              | jumlah_buku | total_harga  | tanggal_transaksi |
+--------------+-----------------------------------------+-------------+--------------+-------------------+
|            4 | Jakarta Sebelum Pagi                    | 1           | Rp85.000,00  | 1 Mei 2026        |
|            5 | Aroma Karsa                             | 1           | Rp160.000,00 | 2 Mei 2026        |
|            6 | Tragedi Pedang Keadilan                 | 1           | Rp90.000,00  | 2 Mei 2026        |
|            7 | Ronggeng Dukuh Paruk, Laut Bercerita    | 2           | Rp175.000,00 | 2 Mei 2026        |
|            8 | Tiga Dalam Kayu                         | 1           | Rp50.000,00  | 2 Mei 2026        |
|            9 | Laskar Pelangi                          | 1           | Rp75.000,00  | 3 Mei 2026        |
|           10 | Teka-Teki Rumah Aneh                    | 1           | Rp60.000,00  | 3 Mei 2026        |
+--------------+-----------------------------------------+-------------+--------------+-------------------+
7 rows in set (0.001 sec)
```

```
MariaDB [bookstore]> delete from buku
    -> where id in(108, 109, 110);
Query OK, 3 rows affected (0.009 sec)
```

```
MariaDB [bookstore]> select * from buku;
+-----+--------------------------+-----------------------+--------------------+--------+
| id  | judul_buku               | nama_penulis          | penerbit           | harga  |
+-----+--------------------------+-----------------------+--------------------+--------+
| 101 | Jakarta Sebelum Pagi     | Ziggy Z               | Grasindo           | 85000  |
| 102 | Teak-Teki Gambar Aneh    | Uketsu                | Grasindo           | 65000  |
| 103 | Tragedi Pedang Keadilan  | Keigo Higashino       | Gramedia           |      0 |
| 104 | Bumi Manusia             | Pramoedya Ananta Toer | Lentera Dipantara  |      0 |
| 105 | Laskar Pelangi           | Andrea Hinata         | Bentang Pustaka    |      0 |
| 106 | Aroma Karsa              | Dee Lestari           | Bentang Pustaka    |      0 |
| 107 | Ronggeng Dukuh Paruk     | Ahmad Tohari          | Gramedia           |      0 |
+-----+--------------------------+-----------------------+--------------------+--------+
7 rows in set (0.001 sec)
```